

FBISE CLASS 9 MATHEMATICS

Chapter 1: Real Numbers

Comprehensive SLO-Based Practice Test

Practice paper designed for annual-examination preparation; not an official FBISE paper.

Name: _____ Roll No.: _____ Date: _____

Assessed Summative SLOs only: M-09-A-04 Apply laws of indices to simplify radical expressions; M-09-A-05 Express a number in scientific notation and vice versa.

Excluded from this test: history of numbers, classification/description of the real-number set, and properties of equality/inequality because these are marked formative in the supplied assessment scheme.

General instruction: Attempt all questions for complete chapter preparation. Assume variables are positive real numbers wherever rational indices/radicals require it.

SECTION A: MULTIPLE CHOICE QUESTIONS

Question 1: Encircle the correct option. (20 MCQs)

1. $a^r \cdot a^{-s} \div a^s$ is equal to

- (a) a^{r-s} (b) a^{r+2s} (c) a^{r-2s} (d) a^{-r+2s}

2. $\sqrt[n]{ab}$ is equivalent to

- (a) $a^{1/n} + b^{1/n}$ (b) $(ab)^{1/n}$ (c) $(a+b)^{1/n}$ (d) $(ab)^n$

3. $(x^{2/3})^6$ simplifies to

- (a) x^2 (b) x^3 (c) x^4 (d) x^{12}

4. $81^{3/4}$ equals

- (a) 9 (b) 27 (c) 81 (d) 243

5. $16^{-3/4}$ equals

- (a) 8 (b) $\frac{1}{8}$ (c) $\frac{1}{4}$ (d) $\frac{1}{16}$

6. $(\frac{16}{25})^{-1/2}$ equals

- (a) $\frac{4}{5}$ (b) $\frac{5}{4}$ (c) $\frac{16}{25}$ (d) $\frac{25}{16}$

7. $\sqrt[3]{216}$ equals

- (a) 3 (b) 6 (c) 12 (d) 36

8. $\frac{\sqrt[4]{x^7}}{\sqrt[4]{x^3}}$ simplifies to

- (a) x (b) x^2 (c) $\sqrt{x}$ (d) $\sqrt[4]{x}$
9. For $a \neq 0$, $a^0 =$
- (a) 0 (b) 1 (c) a (d) -1
10. $(p^3q^{-2})^{-1}$ is
- (a) $\frac{p^3}{q^2}$ (b) $\frac{q^2}{p^3}$ (c) $p^{-3}q^{-2}$ (d) p^3q^2
11. $(\sqrt[3]{5})^6$ equals
- (a) 5 (b) 15 (c) 25 (d) 125
12. $32^{2/5}$ equals
- (a) 2 (b) 4 (c) 8 (d) 16
13. The standard scientific notation of 62,500 is
- (a) 6.25×10^4 (c) 0.625×10^5
 (b) 62.5×10^3 (d) 6.25×10^5
14. 0.00058 in standard scientific notation is
- (a) 5.8×10^{-3} (c) 58×10^{-5}
 (b) 5.8×10^{-4} (d) 0.58×10^{-3}
15. 7.24×10^5 in ordinary form is
- (a) 72,400 (b) 724,000 (c) 7,240,000 (d) 0.000724
16. 8.39×10^7 in ordinary form is
- (a) 8,390,000 (b) 83,900,000 (c) 839,000,000 (d) 0.000000839
17. 0.00406 in standard scientific notation is
- (a) 4.06×10^{-2} (b) 4.06×10^{-3} (c) 4.06×10^3 (d) 0.406×10^{-2}
18. 9.11×10^{-7} in ordinary decimal form is
- (a) 0.000000911 (b) 0.00000911 (c) 0.0000911 (d) 9,100,000
19. Which one is not in standard scientific notation?
- (a) 9.7×10^{-3} (b) 1.04×10^8 (c) 12.4×10^3 (d) 3×10^0
20. The standard scientific notation of 1,002,000 is

(a) 1.002×10^6

(b) 10.02×10^5

(c) 1.002×10^5

(d) 1002×10^3

SECTION B: SHORT QUESTIONS

Questions 2–9: Show essential working.

Q2. Using the product rule for radicals, simplify $\sqrt[3]{6} \times \sqrt[3]{36}$.

Q3. Write $(216)^{-2/3}$ as an equivalent radical expression and simplify completely.

Q4. Use the laws of indices to simplify $\frac{16^{1/5} \cdot 16^{1/4}}{16^{-3/10}}$.

Q5. Write $(\sqrt[4]{10})^8$ in exponential form and simplify.

Q6. Express 73,500,000 in standard scientific notation.

Q7. Express 0.000000482 in standard scientific notation.

Q8. Write 6.307×10^5 in ordinary form.

Q9. Write 9.04×10^{-6} in ordinary decimal form.

SECTION C: LONG QUESTIONS

Questions 10–13: Give complete steps and simplify final answers.

Q10. Use laws of indices/radicals to simplify each expression.

(a) $\frac{64^{2/3} \cdot 8^{-1/3}}{2^{-2}}$

(b) $\left(\frac{x^{1/2}}{y^3}\right)^{-2/3}$

(c) $\sqrt[4]{\frac{a^3}{b^3}} \sqrt[4]{\frac{b^5}{c^3}} \sqrt[4]{\frac{c^4}{a^3}}$

Q11. Prove by using suitable laws of exponents that

$$\left(\frac{x^p}{x^q}\right)^{p+q} \left(\frac{y^q}{y^r}\right)^{q+r} \left(\frac{z^r}{z^p}\right)^{r+p} = x^{p^2-q^2} y^{q^2-r^2} z^{r^2-p^2},$$

where all bases are non-zero and the expressions are defined.

Q12. Scientific notation conversions.

(a) Express each number in standard scientific notation: 0.0000000648, 92,700,000,000, 0.00509.

(b) Write each number in ordinary form: 3.406×10^8 , 7.02×10^{-5} , 1.0005×10^6 .

Q13. Apply scientific notation to the following data. Convert each quantity to the requested form; no unit conversion is required.

- (a) The mean Earth-Sun distance is approximately 149,600,000 km. Write it in standard scientific notation.
- (b) A microorganism is 0.0000042 m long. Write this length in standard scientific notation.
- (c) A measured time is 6.75×10^{-8} s. Write it in ordinary decimal form.
- (d) A current is 0.00000000093 A. Write it in standard scientific notation.
- (e) A data set contains 4.21×10^9 entries. Write this number in ordinary form.

End of Test

Recheck powers, signs of exponents, decimal-point shifts, and the condition $1 \leq a < 10$ in scientific notation.

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